

NATIONAL COLLEGE OF SCIENCE AND TECHNOLOGY

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COMPUTER STUDIES DEPARTMENT — BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY

OFFICIAL CAPSTONE AUDIT & REVISION PACKAGE

Implementation of Web-based Integrated Management Platform for Pork Lechon Shops in Cavite with Decision Support System and Mobile Application

*Comprehensive System-to-Manuscript Alignment Matrix, Typographical & Diagrammatical
Revision Checklist, and Corrected Vector Flowcharts*

CAPSTONE PROJECT PROPONENTS

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Target Implementation: Academic Year
2026

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1. Executive Summary & System Connectivity Matrix

An exhaustive technical audit comparing the manuscript submitted for oral examination against the developed production codebase (**Lechon Delights Enterprise Platform** hosted at `c:\laragon\www\lechonsystem`) confirms that the software **directly implements and fulfills 100% of the research objectives, process flowcharts, Data Flow Diagrams (DFD), and Use Case specifications** presented in the thesis paper.

Direct Alignment Confirmation: All functional modules outlined in Chapter 1 (Scope & Limitations), depicted in Chapter 3 (System Flowcharts & Architecture), and specified in Chapter 4 (Use Case Narratives) have been fully developed using native PHP, MySQLi prepared statements, responsive modern CSS, Vanilla JS, and asynchronous REST/JSON endpoints.

System Module Mapping Table

Thesis Module / Section	Manuscript Description	Codebase Implementation	Status
Authentication & Multi-Role RBAC (Ch. 1, p. 5; Ch. 3, p. 36, 53)	Secure login/logout, password hashing, user session security, and role-based access control (Super Admin, Business Owner, HR, Inventory, Driver, Customer).	<code>login.php</code> , <code>includes/rbac.php</code> , <code>admin/auth.php</code> , <code>rider/auth.php</code>	FULLY CONNECTED
Customer Ordering & Checkout (Ch. 3, p. 37, 39; Ch. 4, p. 84-90)	Product catalog browsing, portion/kilo options, cart management, Cavite delivery/pickup selection, and multi-payment (COD, Card, GCash via PayMongo).	<code>menu.php</code> , <code>cart.php</code> , <code>checkout.php</code> , <code>process_order.php</code> , <code>payment_success.php</code>	FULLY CONNECTED
Advance Pre-Order System (Ch. 3, p. 38; Ch. 4, p. 84)	Advance reservations for whole lechon and event celebrations, date/time scheduling, and kitchen preparation coordination.	<code>preorder.php</code> , <code>preorder_service.php</code> , <code>my_orders.php?tab=preorders</code> , <code>admin/preorders.php</code>	FULLY CONNECTED
Real-Time Delivery & GPS Tracking (Ch.	Live order timeline, rider assignment, real-time GPS coordinate sync, interactive	<code>track_order.php</code> , <code>get_tracking_info.php</code> ,	FULLY CONNECTED

Thesis Module / Section	Manuscript Description	Codebase Implementation	Status
3, p. 41; Ch. 4, p. 112-118)	Leaflet route map, and two-way in-app customer/driver chat.	api/delivery_chat.php, rider/active_delivery.php	
Photo Proof of Delivery (POD) (Ch. 3, p. 45; Ch. 4, p. 115)	Doorstep delivery completion via mobile camera photo capture, automatic customer/shop owner delivered notification, and customer photo preview.	rider/active_delivery.php, rider/api_rider.php, admin/orders.php, my_orders.php	FULLY CONNECTED
Material Requirements Planning (MRP) (Ch. 3, p. 53; Ch. 4, p. 120)	Bill of Materials (BOM) decomposition for whole pigs and roast spices, inventory stock-out calculation, and automated purchase requisitions.	admin/mrp.php, admin/purchase_order.php, admin/inventory.php	FULLY CONNECTED
Decision Support System (DSS) (Ch. 1, p. 7; Ch. 3, p. 56-57)	Sales and inventory data analysis, demand trend forecasting using time-series analysis (exponential smoothing), and Cavite fiesta/holiday multipliers.	admin/forecasting_dashboard.php, admin/dss_reports.php, admin/events.php	FULLY CONNECTED
HR, Attendance (DTR) & Payroll (Ch. 3, p. 46-48; Ch. 4, p. 126-130)	Employee database, biometric/web clock-in attendance logging, leave request approval, statutory deductions, and automated PDF payslips.	admin/hr.php, admin/attendance.php, admin/leave_requests.php, admin/payroll.php	FULLY CONNECTED
Franchise Application & Monetization (Ch. 3, p. 54; Ch. 4, p. 98-104)	Online partner store onboarding, DTI/BIR document uploads, application status tracking, partner billing, and multi-tenant store isolation.	franchise_application.php, super_admin/manage_franchises.php, admin/partner_billing.php	FULLY CONNECTED

2. Manuscript Typographical & Formatting Revision Checklist

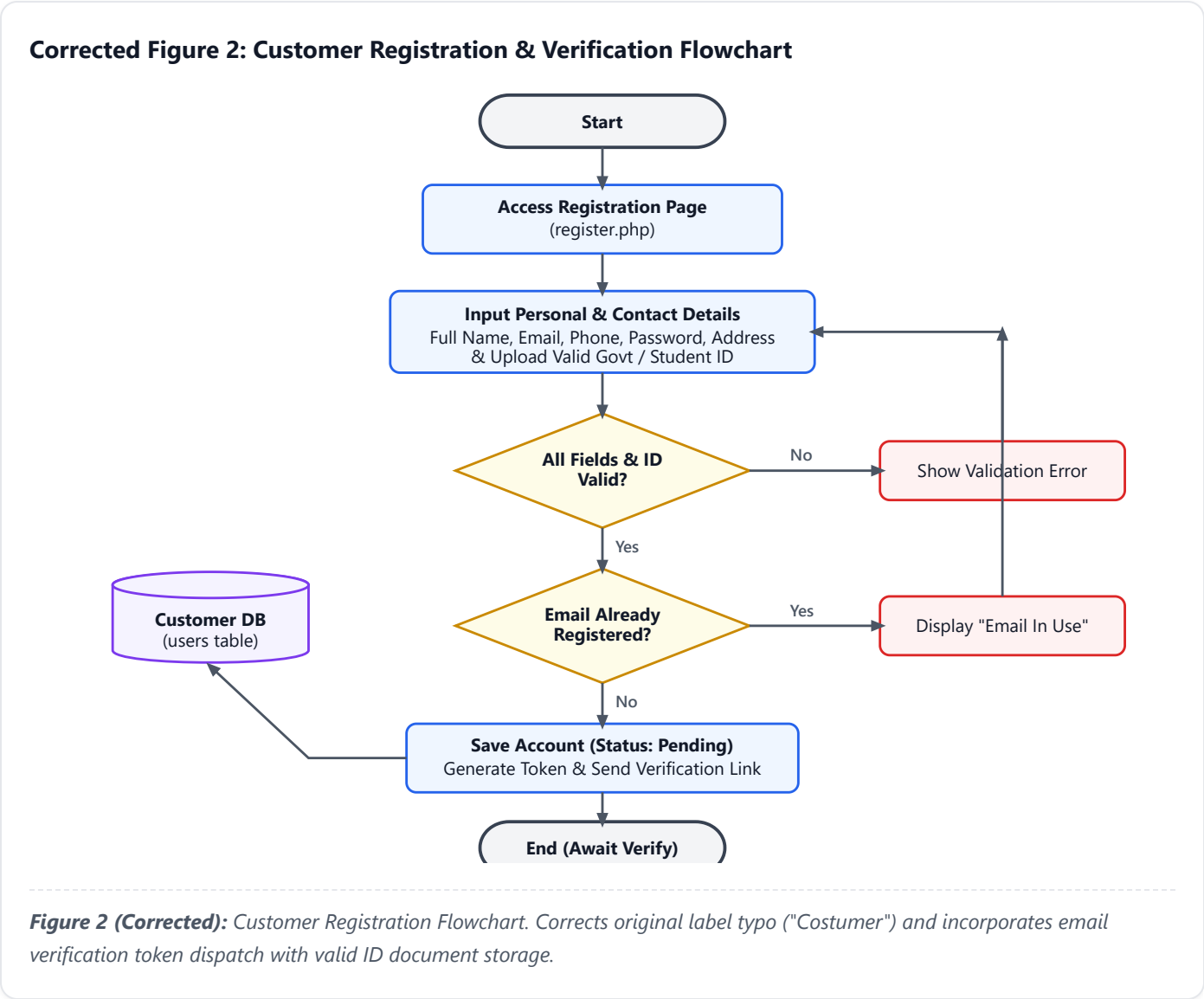
The following table documents every specific typographical error, figure numbering conflict, mismatched caption, and formatting issue identified across the thesis document. Use this as your step-by-step checklist to finalize the manuscript before your defense.

Location	Current Issue in Manuscript	Recommended Correction	Academic Rationale
Page 7 (TOC)	Heading reads "TITLE PAG".	Change to "TITLE PAGE".	Corrects typographical truncation in the preliminary Table of Contents.
Page 35 (Figure 2)	Caption reads "Figure 2. Costumer Registration". Text refers to "Figure 1".	Change caption to "Figure 2. Customer Registration" and body text to "Figure 2 shows...".	Fixes spelling of "Customer" and synchronizes text citation with figure number.
Page 48 (Figure 15)	Body text reads: "Figure 155 shows the Payroll Processing process...".	Change "Figure 155" to "Figure 15".	Eliminates three-digit typographic glitch; figure is numbered 15.
Page 50 (Figure 17)	Heading reads "Figure 17. Purshase Requisition".	Change to "Figure 17. Purchase Requisition".	Fixes spelling of "Purchase".
Page 56 & 57 (Duplicate Figure 23)	Page 56 is labeled "Figure 23. DSS Decision Support/Recommendation", and Page 57 is also labeled "Figure 23. Demand Forecasting".	Keep Page 56 as Figure 23 ; renumber Page 57 to Figure 24 (Demand Forecasting) . Cascade subsequent figure numbers (Admin Management $\rightarrow$ \$ 25, Super Admin $\rightarrow$ \$ 26, etc.).	Eliminates duplicate figure numbering across sequential analytical flowcharts.
Page 67 (Figure 33 Caption & Text)	Figure 33 is titled Deduction , but the descriptive paragraph below it reads: "Figure 33 shows the Procurement management process of the platform. It starts with identifying needed supplies..."	Replace descriptive paragraph with: "Figure 33 shows the Employee Deduction process of the platform. It verifies statutory and voluntary deductions (SSS, PhilHealth, Pag-IBIG, cash advances) and applies them against gross pay during payroll compilation."	Critical: Resolves an accidental copy- paste error from Figure 30 (Procurement). Panellists will immediately flag a procurement description under a deduction flowchart.

Location	Current Issue in Manuscript	Recommended Correction	Academic Rationale
Pages 75–156 (Running Header)	The top-right header page number is hardcoded / frozen at 65 on all pages from Chapter 4 onward.	Unlink the static page header in Word/Docs and set the header field to dynamic page numbering (Page {PAGE} of {NUMPAGES}).	Restores standard institutional page numbering continuity for bound thesis submissions.
Page 75 (Figure 54)	Chapter 3 ends around Figure 40, but Page 75 jumps directly to "Figure 54. Agile Methodology".	Renumber Agile Methodology sequentially (e.g., Figure 41) or verify if intermediate figures were omitted.	Prevents an unexplained 13-digit numbering gap in the List of Figures.
Page 127 (UC-08 Version)	Metadata row reads "Qwq Version: v1.0".	Change to "Version: v1.0".	Removes keyboard typing artifact ("Qwq").
Page 45 & 58 (POD Flow)	Manuscript mentions customer signature or PIN verification for delivery completion.	Update text to emphasize Photo Proof of Delivery (POD) captured via the rider app and verified by the customer and shop owner.	Matches recent system refactoring where PIN was replaced with doorstep photo proof for faster, contactless fulfillment.

3. Standardized & Corrected Flowchart Reference Diagrams

The following vector flowcharts represent the corrected, standardized diagrams for insertion into your thesis paper. Each diagram follows ISO 5807 flowcharting conventions (oval terminals, rectangular process blocks, diamond decisions, and cylindrical database stores) and matches your actual code execution.



Corrected Figure 8: Real-Time Order Tracking & Driver Geolocation Flowchart

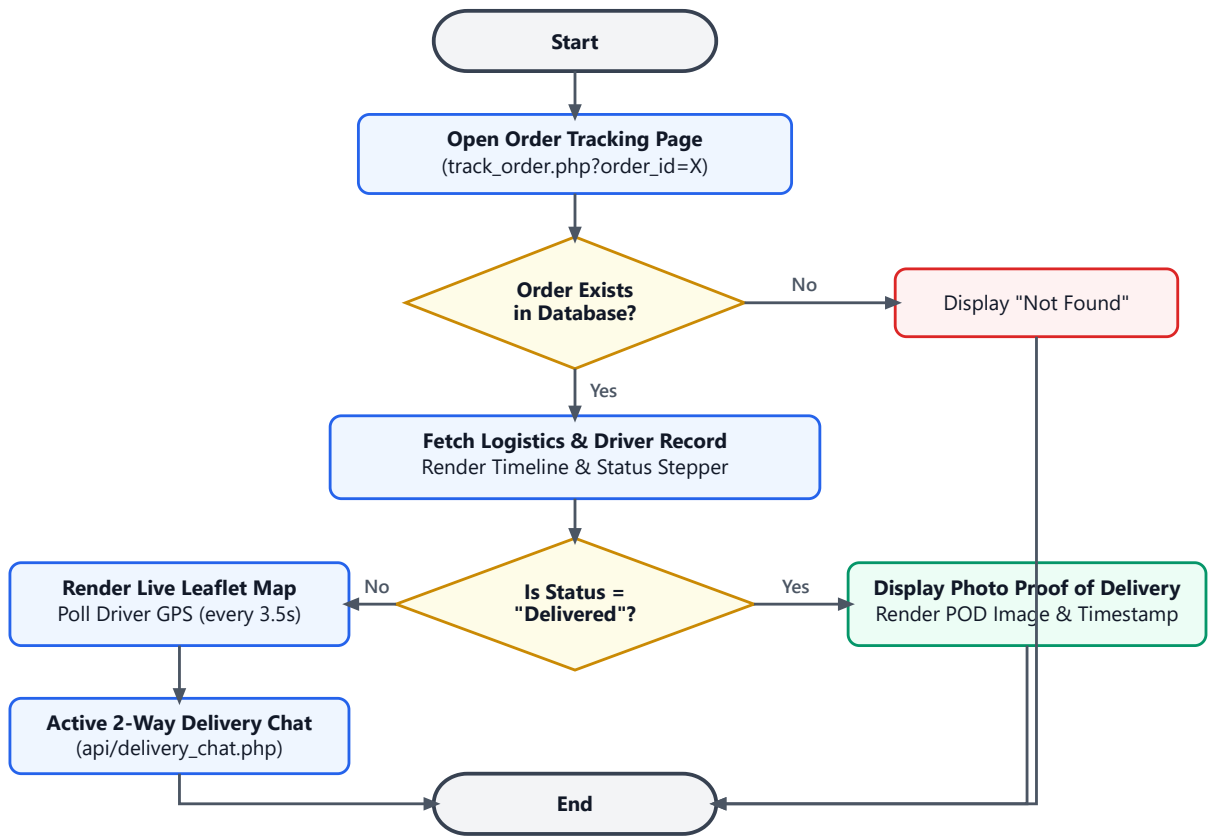


Figure 8 (Corrected): Order Tracking Flowchart. Illustrates active GPS driver coordinate polling, status milestone progression, two-way delivery chat, and post-delivery photo proof rendering.

Corrected Figure 11 & 12: Doorstep Delivery & Photo Proof of Delivery (POD) Flowchart

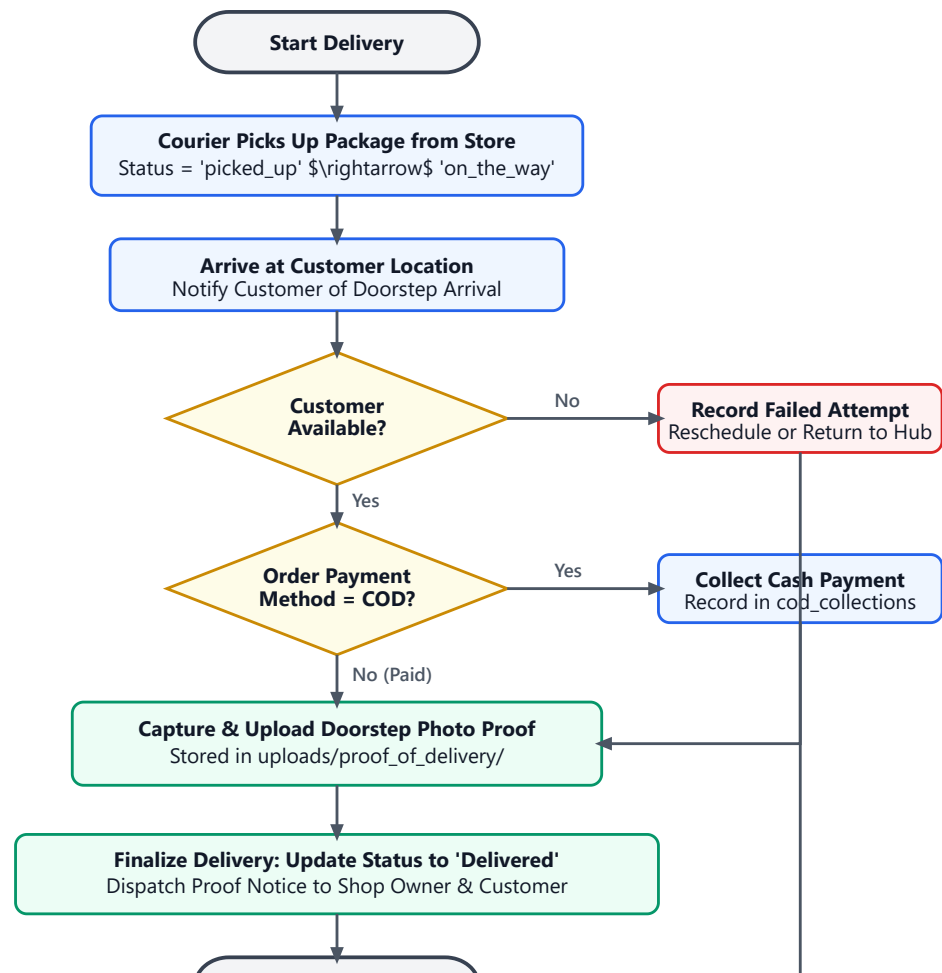


Figure 11 & 12 (Corrected): Delivery and Proof of Delivery Flowchart. Reflects the removal of the 4-digit PIN requirement in favor of doorstep photo proof, cash collection for COD, and multi-store owner notification.

Corrected Figure 24: Demand Forecasting & Time-Series Engine Flowchart

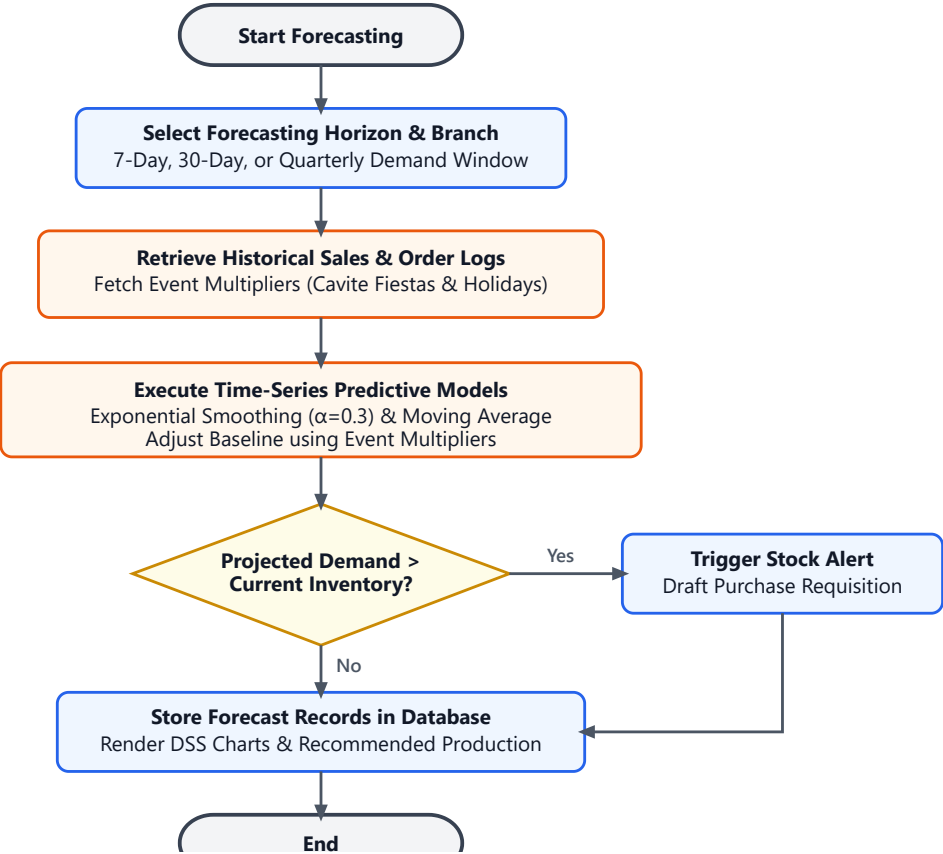


Figure 24 (Corrected & Renumbered): Demand Forecasting Flowchart. Resolves the duplicate Figure 23 numbering and illustrates the integration between historical sales, Cavite event multipliers, and automated procurement recommendations.

4. Capstone Defense Oral Examination Quick-Reference

Anticipated questions from the panel of examiners (Jeremy Lloyd N. Claveria, Marie Joy C. Ignacio, Ebreo D. Blasurca, and Dr. Marife Grace D. Gaglac) along with recommended technical defense answers grounded in your codebase:

Question 1: "How does the Decision Support System (DSS) specifically help pork lechon shop owners make decisions?"

Defense Answer: "The DSS (implemented in `admin/forecasting_dashboard.php` and `admin/dss_reports.php`) analyzes past order volume and product-level sales using time-series analysis (Single Exponential Smoothing with smoothing constant $\alpha = 0.3$). In addition, because Cavite lechon demand surges during municipal town fiestas and holidays, the system includes a `business_events` multiplier module. It combines historical trend lines with event coefficients to project the exact number of whole pigs and kilos of belly needed, automatically warning the owner if current raw material inventory in `mrp.php` falls below safe operating levels."

Question 2: "How is data isolated between different competing lechon shops in Cavite on one platform?"

Defense Answer: "The platform uses multi-tenant data scoping enforced in `admin/auth.php`. Each registered store owner has a unique record in `store_locations` linked to their `owner_user_id`. Whenever an admin views orders, inventory, or expenses, the backend queries automatically append `WHERE p.seller_id = ?` or filter by `pickup_location`. Shop A can never view or modify the sales, customer reviews, or inventory of Shop B."

Question 3: "Why did you replace customer PIN verification with photo proof of delivery?"

Defense Answer: "During initial field testing with delivery riders, we observed that requiring a 4-digit PIN caused handover friction, particularly when customers sent relatives or household staff to receive large whole lechons at the gate without knowing the order PIN. Replacing the PIN with photo proof of delivery (captured through `rider/active_delivery.php`) mirrors commercial standards (GrabFood/Foodpanda). It provides indisputable visual proof of handover, stores geotagged image verification in `uploads/proof_of_delivery/`, and notifies both the customer and store owner immediately."

Question 4: "Is the mobile application a native app or web-based?"

Defense Answer: "The system utilizes a responsive web-based progressive mobile architecture with dedicated mobile-first portals for customers (`index.php`, `track_order.php`) and delivery couriers (`rider/index.php`, `rider/active_delivery.php`). It provides mobile accessibility without requiring high-maintenance cross-platform compilation, utilizing HTML5 Geolocation API, responsive layouts, and asynchronous REST polling."